

Tell me more about:

STATISTICAL SHAPE MODELLING

Your skeleton is unique. Statistical shape modelling helps us predict what it looks like in 3D.



A statistical shape model (SSM) captures how the shape of a bone varies across a population.



The SSM is built from hundreds or thousands of 3D bone scans and learns the most common patterns of anatomical variation.



Statistical methods can then be used to predict the shape of an individual's bones from a limited set of measurements.



Bone shape varies between individuals, due to body size, age, sex, ethnicity, or activity history. SSMs help researchers understand how anatomy differs across people and populations.

Personalised bone models help researchers study how people move. They can be used in musculoskeletal simulations to estimate joint mechanics. Rapid anatomy prediction can reduce the need for expensive medical imaging.

How does this demo work?

A few measurements from your body are used to estimate the shape of your bones. The SSM compares these measurements to patterns learned from many image-based models. Your 3D model is then generated in seconds.



Waipapa
Taumata Rau
University
of Auckland

Auckland
Bioengineering
Institute
www.abi.auckland.ac.nz